

CLINICAL BENCHMARK & KAGGLE-CURATED MEDICAL REFERENCE

Respiratory Diseases & Therapeutics Handbook

An Open Clinical Knowledge Base of Major Pulmonary Diseases, Kaggle-Standard Feature Profiles, Diagnostic Presentation Triads, and Multi-Tiered Pharmacotherapy Protocols Ground-Truthed with WHO & NICE Standards.

Dataset Source: Kaggle Respiratory Disease Benchmarks & Clinical Open Data

Guideline Standards: WHO Essential Guidelines & NICE Clinical Guidelines (UK)

Ontologies: ICD-11 & SNOMED CT Clinical Concept Identifiers

Target Architecture: Tabular Features (Age, Gender, Smoking, Biomarkers) + Symptom Triad inputs $\rightarrow$ Disease (Y_1) & Therapeutics (Y_2)

Curated for AI Researchers, Data Scientists & Healthcare NLP

Standard Medical Edition

Clinical Monograph Reference Guide

Structured disease definitions, demographic profiles, diagnostic criteria, and evidence-based treatments

OBSTRUCTIVE AIRWAY DISEASE

Asthma & Bronchial Asthma

ICD-11: CA23 / CA23.0 SNOMED: 195967001

1. PATHOPHYSIOLOGY & DEFINITION

Chronic airway inflammation mediated by eosinophils and Th2 cytokines causing reversible bronchoconstriction, bronchial hyperresponsiveness, and airway remodeling.

2. DATASET PROFILE & BIOMARKERS (KAGGLE BENCHMARK FEATURES)

Typical Age Range	5 - 45 years	Gender Distribution	M:F ~ 1:1.2
Smoking Impact	Moderate trigger (High exacerbation risk)	Diagnostic Biomarkers	FeNO >35 ppb, Spirometry: FEV1/FVC < 0.70 (Reversibility >12% & 200mL)

3. CLINICAL PRESENTATION TRIAD (\$X_1, X_2, X_3\$)

- Wheezing on expiration
- Shortness of breath (Dyspnea, nocturnal worsening)
- Chest tightness & dry irritant cough

4. EVIDENCED TREATMENT PROTOCOLS (TARGET \$\$)

First Line: Inhaled Corticosteroid (ICS) + Formoterol (MART / AIR strategy)
Second Line: Add LAMA (Tiotropium) or Leukotriene Receptor Antagonist (Montelukast)
Acute Rescue: High-dose SABA (Salbutamol 100mcg) via spacer + Oral Prednisolone (40mg x 5 days)

NICE Guideline: NICE NG245: Asthma: diagnosis, monitoring and chronic management
WHO Standard: WHO Package of Essential NCD Interventions (WHO PEN) Protocol 2

Chronic Obstructive Pulmonary Disease (COPD & Emphysema)

ICD-11: CA22 / CA21

SNOMED: 13645005 / 87433001

1. PATHOPHYSIOLOGY & DEFINITION

Small airway inflammation and parenchymal destruction (emphysema) leading to permanent airflow limitation, loss of elastic recoil, and air trapping.

2. DATASET PROFILE & BIOMARKERS (KAGGLE BENCHMARK FEATURES)

Typical Age Range	40+ years	Gender Distribution	M:F ~ 1.3:1
Smoking Impact	Critical (>85% attributable risk, >10 pack-years)	Diagnostic Biomarkers	Post-bronchodilator FEV1/ FVC < 0.70, Alpha-1 Antitrypsin deficiency screen

3. CLINICAL PRESENTATION TRIAD (\$X_1, X_2, X_3\$)

- Progressive exertional breathlessness
- Chronic productive cough with sputum
- Fatigue, wheezing, and frequent winter chest infections

4. EVIDENCED TREATMENT PROTOCOLS (TARGET \$\$)

First Line: Smoking cessation + Dual Bronchodilation (LAMA + LABA inhaler)
Second Line: Triple Therapy (ICS + LAMA + LABA) if blood eosinophils >=300 cells/uL or frequent exacerbations
Supportive: Pulmonary Rehabilitation, Long-Term Oxygen Therapy (LTOT if PaO2 <7.3 kPa), Vaccinations (Influenza, Pneumococcal, RSV)

NICE Guideline: NICE NG115: COPD in over 16s: diagnosis and management

WHO Standard: WHO Global Strategy for Diagnosis, Management, and Prevention of COPD (GOLD 2026)

Pneumonia (Community-Acquired & Atypical)

ICD-11: CA40

SNOMED: 233604007

1. PATHOPHYSIOLOGY & DEFINITION

Microbial alveolar invasion triggering acute exudative inflammation, alveolar filling with neutrophilic infiltrate and fibrin, impairing gas exchange.

2. DATASET PROFILE & BIOMARKERS (KAGGLE BENCHMARK FEATURES)

Typical Age Range	All ages (Bimodal: <5 years & >65 years)	Gender Distribution	Equal
Smoking Impact	High predisposing factor	Diagnostic Biomarkers	C-Reactive Protein (CRP >100 mg/L), Procalcitonin, Consolidation on Chest X-Ray / CT

3. CLINICAL PRESENTATION TRIAD (\$X_1, X_2, X_3\$)

- High fever, rigors, and profuse chills
- Productive cough with purulent/rusty sputum
- Pleuritic chest pain (sharp, worse on deep inspiration) & tachypnea

4. EVIDENCED TREATMENT PROTOCOLS (TARGET \$\$)

First Line: Amoxicillin (500mg-1g TDS) or Doxycycline/Clarithromycin for 5 days (Low severity CRB65=0)

Hospital Mod Severe: Co-amoxiclav + Clarithromycin (IV/Oral) or Ceftriaxone (CRB65 >= 2)

Viral Fungal: Oseltamivir for Influenza viral pneumonia; Voriconazole/Amphotericin B for fungal

NICE Guideline: NICE CG191: Pneumonia in adults: diagnosis and management

WHO Standard: WHO Model Formulary & Integrated Management of Adolescent and Adult Illness (IMAI)

Lung Cancer (NSCLC & SCLC)

ICD-11: 2C25 SNOMED: 363358000

1. PATHOPHYSIOLOGY & DEFINITION

Malignant transformation of bronchial or alveolar epithelium characterized by genomic instability, immune evasion, neoangiogenesis, and distant metastases.

2. DATASET PROFILE & BIOMARKERS (KAGGLE BENCHMARK FEATURES)

Typical Age Range	55 - 80+ years	Gender Distribution	M:F ~ 1.2:1 (Rising in females)
Smoking Impact	Highest risk factor (>85% cases), Radon, Asbestos	Diagnostic Biomarkers	Low-dose CT scan, PET-CT, EGFR/ALK/KRAS/PD-L1 molecular profiling

3. CLINICAL PRESENTATION TRIAD (\$X_1, X_2, X_3\$)

- Persistent unresolving cough (>3 weeks)
- Hemoptysis (coughing up frank blood or blood-streaked sputum)
- Unexplained constitutional weight loss, hoarseness, and chest/shoulder pain

4. EVIDENCED TREATMENT PROTOCOLS (TARGET \$Y\$)

Early Stage: Anatomic surgical resection (Lobectomy via VATS/Robotic) +/- Adjuvant Chemotherapy
Locally Advanced: Concurrent Chemoradiotherapy +/- Durvalumab Consolidation
Metastatic: Targeted TKIs (Osimertinib for EGFR, Alectinib for ALK) or Pembrolizumab (anti-PD-1) + Chemotherapy

NICE Guideline: NICE NG122: Lung cancer: diagnosis and management
WHO Standard: WHO Classification of Tumours: Thoracic Tumours & Global Cancer Guidelines

Pulmonary Embolism (PE)

ICD-11: BD01 SNOMED: 59282003

1. PATHOPHYSIOLOGY & DEFINITION

Thrombus occlusion of pulmonary arteries elevating right ventricular afterload, inducing acute cor pulmonale, severe V/Q mismatch, and systemic hypoperfusion.

2. DATASET PROFILE & BIOMARKERS (KAGGLE BENCHMARK FEATURES)

Typical Age Range	30 - 85 years	Gender Distribution	Equal (Higher in females during pregnancy/OCP use)
Smoking Impact	Synergistic with hypercoagulable states	Diagnostic Biomarkers	D-Dimer (Age-adjusted), CT Pulmonary Angiography (CTPA), PESI Risk Score

3. CLINICAL PRESENTATION TRIAD (\$X_1, X_2, X_3\$)

- Sudden-onset unexplained dyspnea
- Acute pleuritic chest pain and tachycardia (>100 bpm)
- Hemoptysis, dizziness/syncope, and apprehension/anxiety

4. EVIDENCED TREATMENT PROTOCOLS (TARGET \$Y\$)

Hemodynamically Stable: Direct Oral Anticoagulants (Apixaban, Rivaroxaban) or LMWH for >=3-6 months
Massive High Risk: Systemic Thrombolysis (Alteplase 100mg IV over 2 hrs) or Catheter-directed thrombectomy
Supportive: High-flow oxygen, fluid restriction/vasopressors (Norepinephrine) to support RV function

NICE Guideline: NICE NG158: Venous thromboembolic diseases: diagnosis, management and thrombophilia
WHO Standard: WHO Essential Clinical Care in Shock and Vascular Thrombosis Guidelines

Idiopathic Pulmonary Fibrosis (IPF)

ICD-11: CB03.0

SNOMED: 266395000

1. PATHOPHYSIOLOGY & DEFINITION

Aberrant epithelial-mesenchymal repair in genetically susceptible individuals causing unrelenting fibroblast activation, collagen deposition, and alveolar architecture collapse.

2. DATASET PROFILE & BIOMARKERS (KAGGLE BENCHMARK FEATURES)

Typical Age Range	60 - 80 years	Gender Distribution	M:F ~ 2:1 (Male predominant)
Smoking Impact	Strong associated history	Diagnostic Biomarkers	High-Resolution CT (UIP pattern: subpleural basal honeycombing), FVC decline

3. CLINICAL PRESENTATION TRIAD (\$X_1, X_2, X_3\$)

- Insidious, progressive exertional breathlessness
- Chronic dry, hacking cough
- Bilateral dry 'Velcro' inspiratory crackles & digital clubbing

4. EVIDENCED TREATMENT PROTOCOLS (TARGET \$Y\$)

Antifibrotic Therapy: Pirfenidone (TID) or Nintedanib (150mg BD) to slow FVC decline

Supportive: Ambulatory and nocturnal Oxygen therapy, Pulmonary rehabilitation

Definitive: Early referral for Single/Bilateral Lung Transplantation in eligible candidates

NICE Guideline: NICE TA379 & TA504: Pirfenidone and Nintedanib for IPF; NICE CG163

WHO Standard: WHO / ATS / ERS Clinical Practice Guidelines for Interstitial Lung Disease

Pulmonary Arterial Hypertension (PAH)

ICD-11: BB01 SNOMED: 70995007

1. PATHOPHYSIOLOGY & DEFINITION

Progressive remodeling and obliteration of precapillary pulmonary arterioles resulting in increased pulmonary vascular resistance, right heart hypertrophy, and heart failure.

2. DATASET PROFILE & BIOMARKERS (KAGGLE BENCHMARK FEATURES)

Typical Age Range	30 - 65 years	Gender Distribution	F:M ~ 2.5:1 (Female predominant in Group 1)
Smoking Impact	Modulator of underlying lung vascular tone	Diagnostic Biomarkers	mPAP >20 mmHg via Right Heart Catheterization, NT-proBNP, Echocardiogram RVSP

3. CLINICAL PRESENTATION TRIAD (\$X_1, X_2, X_3\$)

- Shortness of breath on exertion with profound fatigue
- Exertional lightheadedness, presyncope, or syncope
- Peripheral edema (bilateral ankle/leg swelling) & elevated JVP

4. EVIDENCED TREATMENT PROTOCOLS (TARGET \$Y\$)

Combination Vasodilators: Endothelin Receptor Antagonist (Macitentan/Ambrisentan) + PDE-5 Inhibitor (Tadalafil)
Advanced Prostacyclin: Inhaled/IV Epoprostenol or Selexipag for high-risk functional class
Adjuvant: Diuretics (Furosemide/Spironolactone) for right heart congestion, supervised exercise

NICE Guideline: NICE Evidence Summary ESU25 & Specialist Pulmonary Vascular Standards
WHO Standard: WHO Pulmonary Hypertension Group 1-5 Classification & 2026 ESC/ERS Guidelines

Tuberculosis (Pulmonary TB)

ICD-11: 1B10 SNOMED: 56717001

1. PATHOPHYSIOLOGY & DEFINITION

Inhalation of droplet nuclei containing Mycobacterium tuberculosis, causing alveolar macrophage phagocytosis, caseating granuloma necrosis, and tissue destruction.

2. DATASET PROFILE & BIOMARKERS (KAGGLE BENCHMARK FEATURES)

Typical Age Range	15 - 55+ years	Gender Distribution	M:F ~ 1.5:1
Smoking Impact	Increases progression risk by 2-fold	Diagnostic Biomarkers	GeneXpert MTB/RIF (NAAT), Sputum Smear (Ziehl-Neelsen), Upper-lobe Cavitation on CXR

3. CLINICAL PRESENTATION TRIAD (\$X_1, X_2, X_3\$)

- Persistent productive cough lasting >2-3 weeks
- Drenching night sweats & low-grade evening fever
- Progressive unexplained weight loss, anorexia, and hemoptysis

4. EVIDENCED TREATMENT PROTOCOLS (TARGET \$Y\$)

Intensive Phase: Rifampicin (R) + Isoniazid (H) + Pyrazinamide (Z) + Ethambutol (E) daily x 2 months
Continuation Phase: Rifampicin (R) + Isoniazid (H) daily x 4 months (Total 6 months)
Drug Resistant Mdr: Bedaquiline + Pretomanid + Linezolid + Moxifloxacin (BPaLM regimen for 6 months)

NICE Guideline: NICE NG33: Tuberculosis: prevention, diagnosis and management
WHO Standard: WHO Consolidated Guidelines on Tuberculosis: Module 4 - Drug-Susceptible & MDR-TB Treatment